

Speaker Cabinets



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One of the questions I get asked frequently is how to design a speaker cab. I'm not the best source of info in the world for this, so luckily Michael Sabolish has come to the rescue with an excellent write up for you all.

Building and Designing Speaker Cabinets for Guitar and Bass

Building your own speaker cabinets can be a challenging task, however the rewards, both financially and personally, can be great. Perhaps you just aren't pleased with the offerings from commercial companies, or have distinct requirements or tastes that don't line up with what's available for sale at your local music store. Or perhaps you just enjoy woodworking as a pastime...regardless of the reason it is the intention of this article to assist inexperienced cabinet designers to design and construct a speaker cabinet for use with their guitar or bass.

Design Parameters

Before embarking on this task, it is important to decide what you are looking for out of this project. Some people wish to save money, and refuse to pay the inflated prices for a Marshall 4x12" stack. Others may want more power handling than what they can buy in the store. Or maybe you want to make an artistic statement...the possibilities are endless!

However having an idea what to expect out of the finished product is important, so you have goals to work towards. If your only concern is saving money, you can rule out vintage Celestion's because they'll set you back around \$75-\$100 a piece (in low quantity). Or if weight is your primary concern, you can avoid certain types of construction materials. Perhaps frequency response is the most important. In general, have the following parameters in mind:

- * Cost
- * Size
- * Power Output

- * Frequency response
- * Weight
- * Appearance

The theorem that says "There's No Free Lunch" absolutely applies with regards to speaker design. If it was possible to design a speaker cabinet that weighed a pound, cost \$1, had a response of 1Hz to 50 kHz, and could handle a zillion watts, then someone would have done it (and been rich!). In reality, a lot of these variables are mutually exclusive: if you want increased frequency response, the power output will decrease, and the cabinet will be larger (all other things held equal). If you want maximum power output, be prepared to pay more for the drivers and sacrifice some frequency response. Etc, Etc. Keep these things in mind as you read over the next sections.

A Note on Speaker Selection

Don't skimp too much on the speakers: they'll dictate the volume, durability, and overall tonality of the cabinet. However you can have all of the above qualities without spending a fortune.

A word to the wise: if you absolutely MUST have Celestions, then by all means buy and use them. However, like most older things in the music industry, they have a "magic" premium attached to them, and hence you must pay more to get them.

I have found that you can get speakers from well-known companies like Eminence, Dayton, and JBL for MUCH less money, and they sound just as good.

Try and stay away from speakers with cloth accordion or treated paper surrounds for any designs other than open backed. The surround on these speakers can be a large source of air leaks in sealed enclosures, which results in lowered efficiency.

Also look for the highest SPL in dB @ 1W. Remember, each 3-dB increase in efficiency results in twice the apparent sound level at a given power input. Most speakers for instrument use are very high (in the high 90's, and some are over 100) for the wattage they accept.

Guitar Cabinets: Open Back (or infinite baffle)

Open back guitar cabinets are extremely easy to design, and typically require the least amount of construction material to manufacture. They have a very bright and full tone, and make great recording and live cabinets due to the fact that they have wonderful "room filling" properties.

One does not need to worry much about internal volumes, calculations, and other design criterion. Designs are mostly centered around form and weight.

To design these types of cabinets, start with the number of drivers. Most open backed cabinets contain 1, 2 or 4 speakers, but choose any number that meets your needs. Remember that twice the number of drivers doesn't necessarily equate to twice the sound output, due to acoustic coupling and standing wave cancellation in the room. However, more speakers is definitely louder due to the increase in cone surface area.

The design starts with the baffle. Make sure it is big enough to hold the speakers while giving at least an inch or two of clearance on all sides. So for a 2x12" design, the length of the baffle would need to be at least $2 \times 12" + 4 = 28$ inches. The width would need to be at least $12" + 2 = 14$ inches. I usually design open back cabinets to be as physically small as possible, so that I use less wood and the overall finished weight is as little as possible. But the design criterion is up to the designer, and poetic license comes with the job.

Guitar Cabinets: Sealed (or acoustic suspension)

Sealed guitar cabinets are the main staples of the 4x12" stack cabinet manufacturers like Fender and Marshall (what guitar player doesn't want a full stack?). Sealed enclosures can (in general) handle more power, decrease the frequency response (due to a natural built-in rolloff), and increase the design time.

Sealed enclosures are designed by selecting an overall speaker and cabinet Q (called Q_{tc} in the Dickason book). The concept of a system Q should be familiar to anyone who has studied electronics and filtering theory. Q can be thought of as an overall system response, accounting for resonance of the system. In general, systems with a low Q can recover from resonance quickly, and the response deviation band is fairly narrow. In terms of audio properties, lower Q_{tc} values can respond to transient signals better (or more appropriately they have better transient response), at the cost of slightly reduce power handling capabilities. As the Q_{tc} is increased, overall low frequency output power is increased, along with increased power handling abilities.

From my own experimentation, lower Q values have a cleaner tone, whereas higher values result in a darker and warmer tone. For guitar cabinets, good values of Q_{tc} are between 0.707 and 1.0. Anything below 0.707 lacks the punch in the midrange tones, and anything above 1.0 lacks detail and adds a ~2db response step in the low-mid range, which will have to be corrected with external EQ.

Once an overall system Q_{tc} is selected, you can find the necessary volumes using the equations in Dickason's book. When doing guitar cabinets, try and select speakers with appropriate Q_{ts} values that will put the F_3 below the lowest fundamental tone produced by the instrument you are playing. For 6 string guitar, lowest fundamental is the low E string that vibrates at 81 Hz. If your F_3 is below this then you won't experience low end rolloff in the frequency range of your instrument. Careful examination of the equations reveals that in order to get lower F_3 's, you'll need to find drivers with the lowest possible Q_{ts} . Again, you must juggle all of these parameters and decide what is most important.

As a case in point: I was asked to design a full stack for the guitarist in a heavy metal band. I asked him about his preferred requirements, and all I got was "make it damn loud". I also inquired as to what key he played in, and the answer was mostly dropped D.

As a translation into design parameters, he's probably looking for more bass output and power handling: this means a higher Q_{tc} is in order. The F_3 should be below the lowest fundamental, which in this case is about 73 Hz (low D). I also had an idea to try and get the response peak to be in line with the D power chord. In this case, I try and make the F_c (or box resonance frequency) at the lower A string (or about 110 Hz). I ended up selecting a Q_{tc} of 1.0. As a side, note, I managed to build these cabinets for around \$210 a piece, or about \$500 for a full stack! An amazing price difference from commercial offerings.

The result was a cabinet stack tailored to the exact needs of the player. The 2.7 dB peak at 110 Hz added a hard punch to the natural 5th of the D power chord, which sounded incredible. These are the kinds of things you can do with your own designs!

Bass Cabinets

Bass cabinets are a whole other beast. The design parameters are somewhat different, and the approach is very different.

Bass guitars don't produce a lot of mid range frequency energy. Even the highest 4th or 5th harmonics of the highest notes (like the 24th fret of the top G string) don't reach much above 1 kHz. If you want to really be safe, plan on a high end rolloff of about 2 kHz.

A lot of bass cabinets have folded horn tweeters, like those from Eden or SWR. I find that amplifying frequencies this high above the fundamentals results in a tone that is a bit shrill to my tastes, but to each his own. If you are looking to build two way designs (not covered here), Dickason's book goes into great detail on everything from baffle mounting to crossover electrical design.

I find that bass guitar speakers available commercially roll off at way to high a frequency. A lot of the commercially woofers designed for car audio and/or home theatre fit the bill nicely, however it all depends on the type of sound you like. I like a lot of low bass and low midrange punch, which indicates a high end rolloff of 1 kHz or so. Hence most of my designs are tailored to these parameters.

Bass Cabinets: Open back

I don't usually bother with open back bass cabinets: they are relegated to mostly practice amp duty. They lack low frequency content and require huge speakers. Stay away from them unless you are looking for something small to hook to a small 10 watt practice amp.

Bass Cabinets: Sealed

Sealed bass cabinets can be done, however realizing a low enough F_3 is extremely difficult given the majority of woofers available for sale. In order to get decent transient response, you must find drivers with extremely low V_{as} levels. Remember that the low E string on a bass rings out at 41 Hz, and if you want to be able to use a 5 string, the low B rings out at 30 Hz. It is very difficult to realize these F_3 's without huge cabinets, but it can be done. Be very careful of F_3 's here: cone excursion goes up as a square with increasing frequency. So if you end up with a cabinet F_3 of around 60Hz, and you pump a lot of low frequency power into it (say from the low B string at 31Hz), you will most likely exceed cone excursion limits and damage the speakers! Always try and stay within the limits imposed by your design.

Bass Cabinets: Ported (or vented)

Vented bass cabinets yield the lowest F_3 's of all the designs. These designs use one or more vent tubes that radiate most of the low frequency energy as you approach the vent frequency of the design. You can test this with any vented subwoofer: stick an object into one of the ports and you'll notice that the very low frequency energy decreases dramatically.

Vented enclosures are designed not by picking a particular Q_{ts} , but rather by picking a specific alignment. An alignment can be thought of as a set of design criterion which yield a set of responses from the system. The Dickason book goes into great detail on the various specifics on each alignment.

Always be sure to select a port size that is roughly half the diameter of the driver you will be using. This prevents wind noise in the tube, which is kind of like blowing across the opening of a partially filled bottle, and hearing the resonate hum. I like to use 10 inch drivers, with a port tube size of 3 to 4 inches.

One design I finished recently used 4, 10" Dayton drivers, and used a pair of 4" vent tubes approximately 7 inches long each. The cabinet dimensions were roughly 24" by 24" wide and high, and the depth was about 13". Although the cabinet is quite large and heavy, it has a overall F3 of 29 Hz, which allows a five string bass to really push out the low B string with authority. Total cost of this entire enclosure was \$175.

Cabinet Construction and Finishing

The finishing of cabinets can take all of the sound advice from a multitude of woodworking lore and knowledge. There are myriad joint types, finishing styles, and bracing styles. I will show some of the methods that work for me and give good results for low to moderate effort.

Most cabinets are made from plywood or MDF (medium density fiberboard). Don't confuse MDF with particle board: particle board is made from much bigger chunks of wood than MDF. You can tell particle board from MDF just by looking at it: particle board is made from sand sized chunks of wood, whereas MDF looks like pressed wood powder.

A tip: don't use particle board! It was never designed to stand up to a lot of vibration abuse. I have seen 1/2" particle board speaker cabinets self-destruct in higher power situations. My favorite materials of choice are a) birch covered plywood in 3/4" thickness, and b) MDF in 1" thickness. These two materials seem to give the best balance between strength and weight, and are both relatively easy to work with using common woodworking tools.

Using a Circular Saw to make Straight Cuts

Always use a table saw to make straight cuts. Since many table saws cannot handle the 4x8' sheets of plywood I have devised another method using a regular circular saw.

Buy a 4' metal straight edge (I use a contractors level), and two C clamps. Draw a straight line on a test piece of wood, and clamp the straight edge to the wood along this line. Then, take your circular saw and run it along the straight edge. Only make a cut that is a few inches long. Then, measure the distance between the drawn line to the inside edge of the cut. On my circular saw, the offset is 1 5/16". You are now ready to make long straight cuts. When you are cutting out the baffles and sides of your cabinet, subtract this offset from your dimensions, and clamp your straight edge along this line. The straight edge now acts as a guide for the saw to make a perfectly straight cut. And when the offset is added back into the dimensions of the wood you are cutting it comes out the right size.

Buy a high quality saw blade for your circular saw. Sears sells nice ones with carbide or diamond tipped blades. The binding glue used in MDF will quickly heat up and dull most normal blades, and the cuts will begin to have large chunks torn out from the sides yielding jagged edges. These are potential sources of air leaks in vented and sealed enclosures. A good saw blade will last quite a few projects.

Joining and Routing Techniques

My woodworking skills are quite sub-par, so most of my joining is done with butt joints. This is simply placing two pieces of wood together at right angles and gluing. I typically reinforce the joint using 1 1/2" drywall screws down through the top piece into the bottom piece for strength. I have never had a joint failure using this method.

Those who are better at woodworking can make better joints. The best joint of course is a dovetail, where the ends of the two joining pieces are notched in a way so that they mesh together making an incredibly strong joint. One can also use mitre joints or biscuit joints if they have the skill to make it work.

I reinforce the joints from the inside using triangle stock that is 1" on bottom and side. I glue this into place from the inside on all right angle joints. This is probably unnecessary, but if you want to be safe it adds some structural strength to the cabinet.

I always round over all external edges with a rounding bit in a hand held router. I usually pick a radius that is one half the width of the stock I am rounding (e.g. for 1" MDF use a 1/2" round over bit). One may be wondering how I can do this given that I put drywall screws into the joining pieces. Here's the trick: glue all joints with Liquid Nails (this also ensures an airtight bond) as well as the drywall screws. Let this stuff dry for a few hours (better yet overnight), then remove all of the drywall screws. I then round over all of the external edges. Then, I countersink the holes and reinstall the screws. If done properly, you can't tell that this has been done even after being covered.

Making the Circular Speaker Cutouts

This is by far the hardest part of finishing the speaker baffle. Making a good circular cut is quite challenging without the proper tools.

There is always the tried a true method of using a reciprocating saw by hand. If you are good with the saw, this works fine. Go slowly and follow the line. If you measured the width of the speaker frame exactly, allow about 1/16" clearance by making the cutout slightly larger, or do like I do an cut the outside of the line. The driver needs a slight amount of play in the hole so that when the screws holding the speaker to the baffle are tightened, the frame of the speaker does not warp.

If you check with most suppliers of speaker building equipment, they sell a contraption to which you can bolt a router. It has a center alignment pin which screws into the center of the speaker cutout, and you simply drag the attached router in a circle to make the cutout. I borrowed one from a friend and they work excellent, and make a perfect circle for a truly professional job.

Covering the Cabinet

For covering the cabinet, I use the industrial black carpet that is backed by a thick coating of a rubber like material. This covering wears very well, and can hide defects in the wood you are using. It is very easy to apply with 3M spray contact adhesive. It is also relatively inexpensive. Take your time and use pieces which are as big as possible. I used to have problems of the light colored wood showing through the places where the carpet is cut and edged together: I solved this problem by painting every corner side black (you can do this the same time as you paint your baffle wood). This yields invisible seams. I also place small black staples at the seams to ensure that they will never pull up.

If you like natural wood finishes, you have two options. The first is to simply stain the birch veneer that covers birch plywood. If you are careful when you build the cabinet (and use mitred corner joints, as the end grain on butt joints looks terrible stained) this can look every bit as good as any quality home furniture. Wood veneers also work well, especially if you are using very ugly plywood.

You can also use a popular covering called Toulex. It looks like black that has some surface texture applied to it. Crate and Marshall cover their cabinets with this. It wears well and looks good, although the seam work is a lot more difficult to pull off cleanly.

Handles and Other Hardware

Buy high quality strap handles. Good ones will have a metal reinforcement bar inside of them. Use a bolt and T nut to hold them in, because you don't want to have the handle pull out of the cabinet sending your prized cabinet tumbling down a flight of stairs (spoken from experience).

Marshall uses recessed handles that mount to the side of the cabinet. I don't like these because they require a large cutout in the both sides of the cabinet, which reduces the strength of the cabinet and is another potential source of air leaks. To each his own, however. Just make sure that if you buy some, that they are metal (not plastic).

Buy corner reinforcement! Cabinets tend to get beat up with use, and the corners are the weakest areas and the first to cave in. Corner reinforcement spreads impact force along the sides of the cabinet making it less likely that the corner will cave. I have used both metal and plastic ones with good success. If you are building a full stack, there are special ones that interlock and prevent your top slanted cabinet from sliding off the bottom one.

I really like to put high quality casters on my cabinets. You can get 3 inch locking casters which support up to 500 lbs. from Home Depot for about \$5 a piece. Your back will thank you for it. Keep in mind that you only want to put them on the bottom cabinet of your stack, as you don't want the top one rolling off in the middle of a gig (although buying locking casters alleviates this problem).

For grill cloth, there are several options. I like to use commercial grill cloth because it is cheap and easy to use. I have also successfully used screen door wire available at home improvement stores. It comes in a nylon/plastic variety as well as metal. The metal stands up to abuse more than either the plastic variety or cloth, however if it is stretched too tight it will hum and make rattling noises.

I have also been informed that commercial sheet metal companies sell an industrial grate used in making tractors and other heavy equipment. It is very heavy, but can withstand just about any abuse thrown at it (e.g. boots, guitars, and beer bottles). You can buy it by the linear yard, although I'm not sure of the cost. It is very reminiscent of what SWR uses on their Goliath cabinets.

A note on acoustic stuffing: I highly recommend that all sealed and vented enclosures have 50% of their internal volume stuffed with acoustic stuffing. You can buy this by the pound from Part's Express, or you can use the fiberglass insulation made for insulating homes. Be warned: the pink fiberglass is a nose, eye, and throat irritant so take the proper precautions. Stuffing actually increases the apparent volume of the cabinet, and helps compensate for volume lost by driver baskets, handles, crossover components, etc. It also helps break up standing waves inside the cabinet.

Wiring

Use high quality silicon wiring of at least 16 gauge for connecting up the speakers. Make sure it is of the braided type, because it has higher current handling capabilities, and is easier to bend around internal obstructions.

Also use a high quality speaker jack on the back of the cabinet. My favorites are the heavy duty binding post type terminals, where the speaker wire actually fits in a hole in the post, and there is a nut to lock it to the terminal. Another favorite is the locking 1/4" jacks by Neutrix. Whatever you do don't use either a) a regular 1/4" guitar jack, or b) the spring loaded push button type terminals. I absolutely guarantee that these types will pull out during a gig! Also solder all wiring connections. A lot of speakers come equipped with push on connectors, but don't use them. If it vibrates loose you'll have to tear the cabinet apart to get at them again.

Also be sure to seal the terminal with caulk or Liquid Nails to eliminate air leaks.

Also be sure your speakers are phased properly before you button up your cabinet. A quite way to check is to take a 9V battery, and touch the positive and negative leads to the respective speaker leads (e.g. where you plug in your amp). All of the drivers will move to a fixed offset. Make sure they all move the same direction. If one woofer sucks in while the other speakers push out you have it out of phase. Fix it now or the sound quality will suffer.

Also ***BE SURE YOUR OUTPUT IMPEDANCE IS CORRECT!*** Many a would be speaker builder has miswired the speakers and ruined perfectly good amplifiers. Keep in mind that solid state amps and tube amps react very differently to mismatched speaker loads.

A solid state amp will still work with a higher impedance load that it is rated to drive. So if you wire your cabinet as an 8 ohm impedance and your amp is expecting 4 ohm, all that will happen is the output volume will suffer. Due to the fact that solid state have a very low output impedance, if you try to put a 2 ohm load on your amp from a miswired cabinet, it will try to source too much current and something will give (usually the fuse, but maybe your power BJT's?).

Tube amps are a different story. Efficiency is assured by the correct output impedance being reflected onto the primary of the output transformer, which in part determines the amount of current flowing through the output tubes. If you put on a wrong impedance (EITHER higher or lower), the output tubes "see" an impedance mismatch and actually put out LESS power into the cabinet. It can also cause transformer flyback and tube socket arcing leading to a quick death to either the output tubes, cathode resistors, or the output transformer (maybe all!). In short, DON'T mismatch cabinet impedance with a tube amplifier!

For two speaker systems, it's common to get drivers with twice the expected impedance of your output amplifier, and wire them in parallel. So if your amp needs 4 ohms, buy two 8 ohm drivers and wire them in parallel to yield 4 ohms. For four speaker systems, I have found it is best to wire in a series-parallel configuration, where two separate pairs of speakers are wired in series, and the resulting pairs are wired in parallel. In this configuration, buy drivers with the same impedance as you need (for a 8 ohm cabinet, buy four 8 ohm drivers). There are other wiring possibilities as well (such as strict series).

Conclusion

I hope this advice helps out budding speaker builders and allows first the first project to be successful. Start out with a smaller project first, and build up to more challenging projects as your knowledge and skill increase.

My favorite first project to recommend is to redo cheap combo practice amps. My project was the Crate 15W: with a decent speaker in a open back or sealed enclosure the sound difference is incredible. The anemic speakers don't do these little amps justice.

Another great first project is the classic 2x12" Fender extension cabinet copy. Two Eminence speakers in an open-backed cabinet for under \$100.

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